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Quiz 5

NAME: ______________________________

Q.1. Write the general equation of the circle with center $C(h, k)$ and radius r. The following is the equation of a circle in the $x - y$ plane: $x^2 - 4x + y^2 = 5$. With working, find its center and radius (**Hint:** start by completing the squares for the x-terms and y-terms separately).

Q.2. I have a line l in the $x - y$ plane. l is perpendicular to the line $y = 1 - x$ and passes through the point $(-1, 4)$. What is the **slope-intercept** form of l? (**Hint:** start by setting up the point-slope form of l)

Q.3. Find the domain of the function $f(x) = \frac{\sqrt{x+1}}{2x}$. Explain all restrictions on the domain clearly (either with equations/ inequalities or in words). Write your final answer in interval notation.

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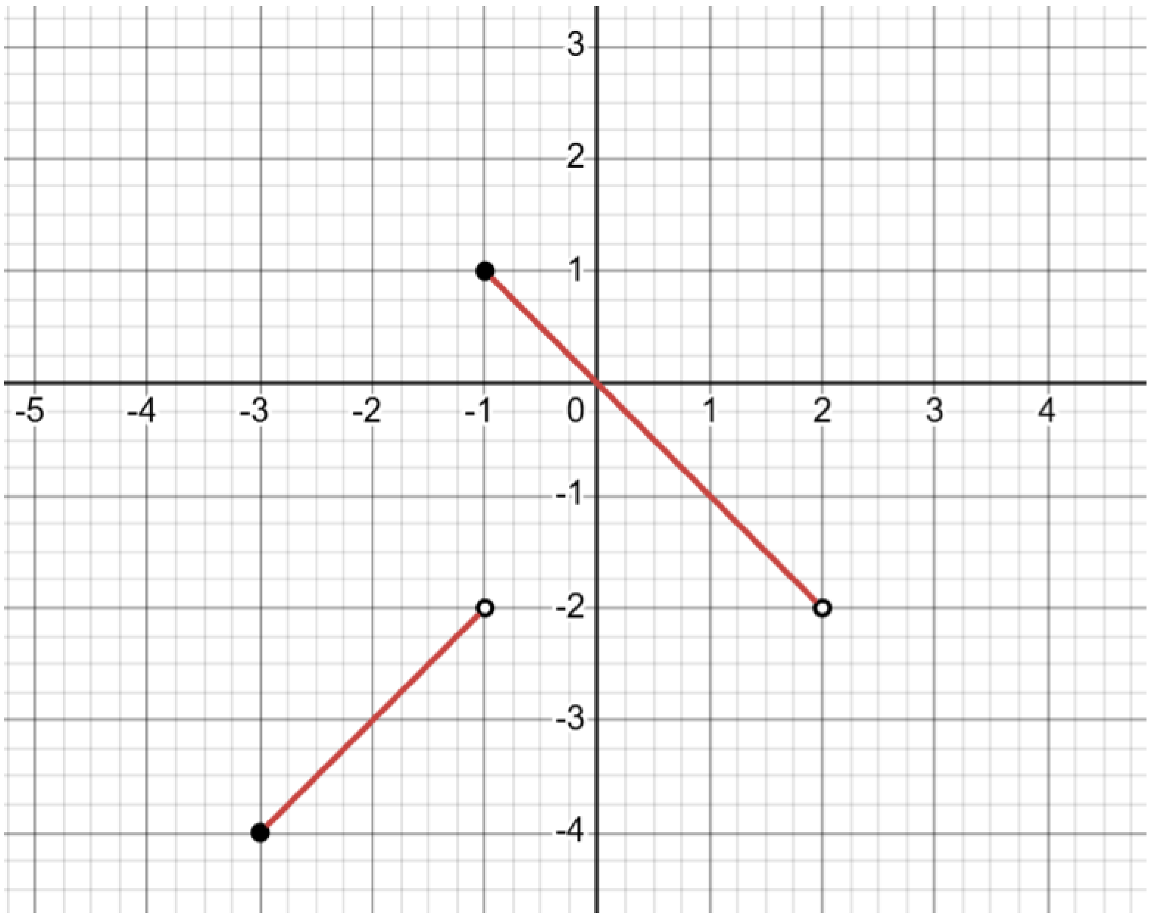
Q.4. The graph of a function $y = f(x)$ is given below:



State the domain and range of $f(x)$ in interval notation. In interval notation, state the intervals on which $f(x)$ is increasing and decreasing, respectively. Clearly mark all your answers.